

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276795

Kind Code

A1

Publication Date

September 04, 2025

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AIRCRAFT PROPULSION SYSTEM

Abstract

A propulsion system for an aircraft includes an air intake, an engine assembly, a turbocompressor, an electrical assembly, and at least one propulsor. The engine assembly includes an engine. The engine includes an air inlet, an exhaust outlet, and an engine output shaft. The turbocompressor assembly includes a turbocompressor. The turbocompressor includes a turbine and a compressor. The turbine and the compressor form a rotational assembly. The rotational assembly includes a shaft, a bladed turbine rotor of the turbine, and a bladed compressor rotor of the compressor. The shaft interconnects the bladed turbine rotor and the bladed compressor rotor. The turbine includes a turbine inlet and a turbine outlet. The turbine inlet is connected in fluid communication with the exhaust outlet. The compressor includes a compressor inlet and a compressor outlet. The compressor inlet is connected in fluid communication with the air intake. The rotational assembly is mechanically independent of the engine output shaft.

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Family ID: 94869181

Appl. No.: 18/591141

Filed: February 29, 2024

Publication Classification

Int. Cl.: B64D27/33 (20240101); B64D33/08 (20060101); B64U10/14 (20230101); B64U20/94 (20230101); B64U20/96 (20230101); B64U50/12 (20230101); B64U50/19 (20230101)

U.S. Cl.:

CPC B64D27/33 (20240101); B64D33/08 (20130101); B64U20/94 (20230101); B64U20/96 (20230101); B64U50/12 (20230101); B64U50/19 (20230101); B64U10/14 (20230101)

Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to a propulsion system for an aircraft.

BACKGROUND OF THE ART

[0002] A propulsion system for an aircraft may be configured, for example, with an engine assembly configured to drive rotation of a propulsor for the aircraft. Various configurations of such aircraft propulsion systems are known in the art. While these known aircraft propulsion systems may be suitable for their intended purposes, there is always room in the art for improvement.

SUMMARY

[0003] It should be understood that any or all of the features or embodiments described herein can be used or combined in any combination with each and every other feature or embodiment described herein unless expressly noted otherwise.

[0004] According to an aspect of the present disclosure, a propulsion system for an aircraft includes an air intake, an engine assembly, a turbocompressor, an electrical assembly, and at least one propulsor. The engine assembly includes an engine and an intercooler. The engine includes an air inlet, an exhaust outlet, and an engine output shaft. The intercooler includes a first heat exchanger side having a first inlet and a first outlet and a second heat exchanger side having a second inlet and a second outlet. The first outlet is connected in fluid communication with the air inlet. The second inlet is connected in fluid communication with the air intake. The turbocompressor assembly includes a turbocompressor. The turbocompressor includes a turbine and a compressor. The turbine and the compressor form a rotational assembly. The rotational assembly includes a shaft, a bladed turbine rotor of the turbine, and a bladed compressor rotor of the compressor. The shaft interconnects the bladed turbine rotor and the bladed compressor rotor. The turbine includes a turbine inlet and a turbine outlet. The turbine inlet is connected in fluid communication with the exhaust outlet. The compressor includes a compressor inlet and a compressor outlet. The compressor inlet is connected in fluid communication with the air intake. The compressor outlet is connected in fluid communication with the first inlet. The rotational assembly is mechanically independent of the engine output shaft. The electrical assembly includes a first electric motor-generator and an electrical distribution system. The first electric motor-generator is operably connected to the engine output shaft. The first electric motor-generator is electrically connected to the electrical distribution system. The at least one propulsor includes an electric motor and a propulsor rotor. The electric motor is electrically connected to the electrical distribution system. The propulsor rotor is operably connected to the electric motor.

[0005] In any of the aspects or embodiments described above and herein, the propulsion system may further include a cooling fan including a fan rotor and an input shaft. The input shaft may be connected to the fan rotor. The input shaft may be operably connected to the rotational assembly. The cooling fan may include a cooling fan inlet and at least one cooling fan outlet. The cooling fan may be connected in fluid communication with the second heat exchanger side to direct ambient air from the air intake through the second heat exchanger side.

[0006] In any of the aspects or embodiments described above and herein, the cooling fan may be connected in fluid communication with and between the air intake and the second heat exchanger side with the cooling fan inlet connected in fluid communication with the air intake and the at least one cooling fan outlet connected in fluid communication with the second inlet.

[0007] In any of the aspects or embodiments described above and herein, the cooling fan may be configured to exhaust ambient air from the air intake through the at least one cooling fan outlet to generate lift for the aircraft.

[0008] In any of the aspects or embodiments described above and herein, the electrical assembly

may include a second electric motor-generator operably connected to the shaft. The second electric motor-generator may be electrically connected to the electrical distribution system.

[0009] In any of the aspects or embodiments described above and herein, the turbocompressor assembly may further include a gearbox. The gearbox may operably connect the shaft, the input shaft, and the second electric motor-generator.

[0010] In any of the aspects or embodiments described above and herein, the propulsion system may further include a cooling fan including an electric motor, a fan rotor, a cooling fan inlet, and at least one cooling fan outlet. The electric motor may be electrically connected to the electrical distribution system. The electric motor may be operably connected to the fan rotor. The cooling fan may be connected in fluid communication with the second heat exchanger side to direct ambient air from the air intake through the second heat exchanger side.

[0011] In any of the aspects or embodiments described above and herein, the cooling fan may be connected in fluid communication with and downstream of the second heat exchanger side. The cooling fan inlet may be connected in fluid communication with the second outlet.

[0012] In any of the aspects or embodiments described above and herein, the engine assembly may include an interburner. The interburner may include an interburner inlet and an interburner outlet. The interburner may be connected in fluid communication with and between the engine and the turbine with the interburner inlet connected in fluid communication with the exhaust outlet and the interburner outlet connected in fluid communication with the turbine inlet.

[0013] In any of the aspects or embodiments described above and herein, the electrical assembly may further include an energy storage device electrically connected to the electrical distribution system.

[0014] In any of the aspects or embodiments described above and herein, each of the at least one propulsor may be connected in fluid communication with the intercooler outlet.

[0015] In any of the aspects or embodiments described above and herein, the engine may be an intermittent combustion engine.

[0016] According to another aspect of the present disclosure, an unmanned aerial vehicle (UAV) includes an air intake, a turbocompressor assembly, a cooling fan, and an engine assembly. The turbocompressor assembly includes a turbocompressor. The turbocompressor includes a turbine and a compressor. The turbine and the compressor form a rotational assembly. The rotational assembly includes a shaft, a bladed turbine rotor of the turbine, and a bladed compressor rotor of the compressor. The shaft interconnects the bladed turbine rotor and the bladed compressor rotor. The turbine includes a turbine inlet and a turbine outlet. The compressor includes a compressor inlet and a compressor outlet. The cooling fan includes a fan rotor and an input shaft. The input shaft is connected to the fan rotor. The input shaft is operably connected to the rotational assembly. The cooling fan includes a cooling fan inlet and at least one cooling fan outlet. The cooling fan inlet is connected in fluid communication with the air intake. The cooling fan outlet is connected in fluid communication with the compressor inlet. The engine assembly includes an engine and an intercooler. The engine includes an air inlet, an exhaust outlet, and an engine output shaft. The exhaust outlet is connected in fluid communication with the turbine inlet. The intercooler includes a first heat exchanger side having a first inlet and a first outlet and a second heat exchanger side having a second inlet and a second outlet. The first inlet is connected in fluid communication with the compressor outlet. The first outlet is connected in fluid communication with the air inlet. The second inlet is connected in fluid communication with the cooling fan outlet.

[0017] In any of the aspects or embodiments described above and herein, the UAV may further include an electrical assembly including a first electric motor-generator, a second electric motor-generator, and an electrical distribution system. The first electric motor-generator may be operably connected to the engine output shaft. The first electric motor-generator may be electrically connected to the electrical distribution system. The second electric motor-generator is operably connected to the rotational assembly. The second electric motor-generator is electrically connected

to the electrical distribution system.

[0018] In any of the aspects or embodiments described above and herein, the input shaft may be operably connected to the rotational assembly and the second electric motor-generator.

[0019] In any of the aspects or embodiments described above and herein, the UAV may further include at least one propulsor including an electric motor and a propulsor rotor. The electric motor may be electrically connected to the electrical distribution system. The propulsor rotor may be operably connected to the electric motor.

[0020] According to another aspect of the present disclosure, a propulsion system for an aircraft includes an air intake, an engine assembly, a turbocompressor assembly, and an electrical assembly. The engine assembly includes an engine, an interburner, and an intercooler. The engine includes an air inlet, an exhaust outlet, and an engine output shaft. The interburner includes an interburner inlet and an interburner outlet. The interburner inlet is connected in fluid communication with the exhaust outlet. The intercooler includes a first heat exchanger side having a first inlet and a first outlet and a second heat exchanger side having a second inlet and a second outlet. The first outlet is connected in fluid communication with the air inlet. The second inlet is connected in fluid communication with the air intake. The turbocompressor assembly includes a turbocompressor. The turbocompressor includes a turbine and a compressor. The turbine and the compressor form a rotational assembly. The rotational assembly includes a shaft, a bladed turbine rotor of the turbine, and a bladed compressor rotor of the compressor. The shaft interconnects the bladed turbine rotor and the bladed compressor rotor. The turbine includes a turbine inlet and a turbine outlet. The turbine inlet is connected in fluid communication with the interburner outlet. The compressor includes a compressor inlet and a compressor outlet. The compressor inlet is connected in fluid communication with the air intake. The compressor outlet is connected in fluid communication with the first inlet. The rotational assembly is mechanically independent of the engine output shaft. The electrical assembly includes a first electric motor-generator, a second electric motor-generator, and an electrical distribution system. The first electric motor-generator is operably connected to the engine output shaft. The first electric motor-generator is electrically connected to the electrical distribution system. The second electric motor-generator is operably connected to the rotational assembly. The second electric motor-generator is electrically connected to the electrical distribution system.

[0021] In any of the aspects or embodiments described above and herein, the propulsion system may further include a cooling fan including a fan rotor and an input shaft. The input shaft may be connected to the fan rotor. The input shaft may be operably connected to the rotational assembly. The cooling fan may include a cooling fan inlet and at least one cooling fan outlet. The cooling fan inlet may be connected in fluid communication with the air intake. The at least one cooling fan outlet may be connected in fluid communication with the compressor inlet and the second inlet.

[0022] In any of the aspects or embodiments described above and herein, the cooling fan may be configured to exhaust ambient air from the air intake through the at least one cooling fan outlet to generate lift for the aircraft.

[0023] In any of the aspects or embodiments described above and herein, the propulsion system may further include an engine exhaust diverter assembly connected in fluid communication with and between the exhaust outlet, the interburner inlet, and the turbine inlet. The engine exhaust diverter assembly may be selectively configurable in a first condition or a second condition: in the first condition, the engine exhaust diverter subassembly may connect the exhaust outlet in fluid communication with the interburner inlet and, in the second condition, the engine exhaust diverter subassembly may connect the exhaust outlet in fluid communication with the turbine inlet.

[0024] The present disclosure, and all its aspects, embodiments and advantages associated therewith will become more readily apparent in view of the detailed description provided below, including the accompanying drawings.

Description

DESCRIPTION OF THE DRAWINGS

[0025] FIG. **1** illustrates a perspective view of an aircraft including a propulsion system, in accordance with one or more embodiments of the present disclosure.

[0026] FIG. **2** schematically illustrates a propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

[0027] FIG. **3** schematically illustrates another propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

[0028] FIG. **4** schematically illustrates another propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

[0029] FIG. **5** schematically illustrates another propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

[0030] FIG. **6** schematically illustrates another propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

[0031] FIG. **7** schematically illustrates another propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

[0032] FIG. **8** schematically illustrates a cutaway, side view of a propulsor for an aircraft propulsion system, in accordance with one or more embodiments of the present disclosure.

[0033] FIG. **9** schematically illustrates a portion of another propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

[0034] FIG. **10** schematically illustrates a portion of another propulsion system for an aircraft, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0035] FIG. **1** illustrates an aircraft **1000** including a propulsion system **20**. Briefly, the aircraft **1000** may be an unmanned aerial vehicle (UAV, e.g., a drone). For example, the aircraft **1000** may be configured as a quadcopter (e.g., as shown in FIG. **1**) or another UAV configuration. Alternatively, the aircraft **1000** may be a fixed-wing aircraft (e.g., an airplane), a rotary-wing aircraft (e.g., a helicopter), a tilt-rotor aircraft, a tilt-wing aircraft, or another aerial vehicle. The propulsion system **20** is configured to facilitate propulsion and electrical power generation for the aircraft **1000**.

[0036] FIG. **2** schematically illustrates the propulsion system **20**. The propulsion system **20** of FIG. **2** includes an engine assembly **22**, a turbocompressor **24**, an electrical assembly **26**, an air intake **28**, a cooling fan **30**, and one or more propulsors **32** (e.g., vertical take-off and landing (VTOL) propulsors).

[0037] The engine assembly **22** includes an engine **34**, a gearbox **36**, an interburner **38**, and an intercooler **40**.

[0038] The engine **34** is an internal combustion engine configured to use intermittent combustion during operation. In other words, the engine **34** may be an engine configuration other than a gas turbine engine configuration. For example, the engine **34** may be a reciprocating engine such as, but not limited to, a piston engine or a rotary engine (e.g., a Wankel engine). The engine **34** includes an engine output shaft **42**, an air inlet **44**, and an exhaust outlet **46**. The engine **34** is configured to drive rotation of the engine output shaft **42**. The engine output shaft **42** is operably connected (e.g., coupled) to the gearbox **36**. The engine **34** is configured to receive compressed air from the intercooler **40** at the air inlet **44** and direct combustion gas (e.g., exhaust gas) to the interburner **38** at the exhaust outlet **46**.

[0039] The gearbox **32** includes at least one gear assembly **48** operably connected (e.g., coupled) to the engine output shaft **42**. The gear assembly **48** may include an epicyclic gear assembly (e.g., a planetary gear assembly). The present disclosure, however, is not limited to any particular gear

configuration for the gear assembly **48**.

[0040] The interburner **38** forms a combustion chamber in which the combustion gas from the engine **34** is mixed and burned with fuel. The interburner **38** includes an inlet **50** and an outlet **52**. The inlet **50** is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the exhaust outlet **46** to receive combustion gas from the engine **34**. The interburner **38** is configured to direct combustion gas to the turbocompressor **24** at the outlet **52**.

[0041] The intercooler **40** is configured as a heat exchanger for cooling the compressed air directed to the air inlet **44** of the engine **34**. The intercooler **40** includes a first inlet **54** and a first outlet **56** for a primary heat exchanger side of the intercooler **40**. The intercooler **40** includes a second inlet **58** and a second outlet **60** for a secondary heat exchanger side of the intercooler **40**. The intercooler **40** is configured to receive compressed air from the turbocompressor **24** at the first inlet **54**. The first outlet **56** is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the air inlet **44**. The compressed air is cooled by heat transfer from the primary heat exchanger side to the secondary heat exchanger side, and directed to the air inlet **44** of the engine **34** from the first outlet **56**. The second inlet **58** is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the air intake **28** and/or the cooling fan **30**. For example, the intercooler **40** is configured to receive cooling air (e.g., ambient air) from the cooling fan **30** at the second inlet **58**. The intercooler **40** is configured to exhaust this cooling air from the propulsion system **20** through the second outlet **60**.

[0042] Turbocompressor assembly **24** of FIG. 2 includes a turbocompressor **62** and a gearbox **64**. The turbocompressor **62** includes a compressor **66** and a turbine **68**. The compressor **66** includes an inlet **70** and an outlet **72**. The inlet **70** is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the air intake **28** and/or the cooling fan **30**. For example, the compressor **66** is configured to receive air (e.g., ambient air) from the cooling fan **30** at the inlet **70**. The outlet **72** is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the intercooler **40** (e.g., the first inlet **54**) to direct compressed air to the intercooler **40**. The turbine **68** includes an inlet **74** and an outlet **76**. The inlet **74** is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the interburner **38** (e.g., the outlet **52**). The turbine **68** is configured to receive the combustion gas from the interburner **38** at the inlet **74**. The turbine **68** is configured to exhaust the combustion gas from the propulsion system **20** through the outlet **76**.

[0043] Components of the compressor **66** and the turbine **68** form a rotational assembly **78** mounted for rotation about a rotational axis (e.g., an axial centerline of the turbocompressor **24**). The rotational assembly **78** includes a shaft **80**, a bladed compressor rotor **82** for the compressor **66**, and a bladed turbine rotor **84** for the turbine **68**. The shaft **80** interconnects the bladed compressor rotor **82** and the bladed turbine rotor **84**.

[0044] The gearbox **64** includes at least one gear assembly **86** operably connected (e.g., coupled) to the rotational assembly **78** (e.g., the shaft **80**). The gear assembly **86** may include an epicyclic gear assembly (e.g., a planetary gear assembly). The present disclosure, however, is not limited to any particular gear configuration for the gear assembly **86**.

[0045] The electrical assembly **26** of FIG. 2 includes a first electric motor-generator **88**, a second electric motor-generator **90**, and an electrical distribution system **92**. The electrical assembly **26** may additionally include an energy storage device **44**.

[0046] The first electric motor-generator **88** is operably connected (e.g., coupled) to the engine **34** (e.g., the engine output shaft **42**) by the gearbox **36** (e.g., the gear assembly **48**). The first electric motor-generator **88** of FIG. 2 includes a drive shaft **96** operably connected to the gear assembly **48**. The gear assembly **48** may be a speed-changing gear assembly configured to drive rotation of the first electric motor-generator **88** (e.g., the drive shaft **96**) at a different rotational speed than the engine output shaft **42**, and vice versa. The first electric motor-generator **88** is electrically connected to the electrical distribution system **92**. The first electric motor-generator **88** is

configured to generate an electrical power output, in response to driving rotation from the engine **34** (e.g., the engine output shaft **42**), and direct the electrical power output to the electrical distribution system **92** (e.g., a generating operational mode). Alternatively, the first electric motor-generator **88** is configured to receive an electrical power input from the electrical distribution system **92** (e.g., from the second electric motor-generator **90**, the energy storage device **94**, or a power source external to the propulsion system **20** or the aircraft **1000** (e.g., ground-based power system or an on-board power system of the aircraft **1000**) and drive or assist rotation of the engine output shaft **42** through the gear assembly **48** (e.g., a motoring operational mode). The electrical power input and/or output of the first electric motor-generator **88** may be in the form of an alternating current (AC) or a direct current (DC), and the present disclosure is not limited to any particular configuration of the first electric motor-generator **88**.

[0047] The second electric motor-generator **90** is operably connected (e.g., coupled) to the rotational assembly **78** (e.g., shaft **80**) by the gearbox **64** (e.g., the gear assembly **86**). The second electric motor-generator **90** of FIG. 2 includes a drive shaft **98** operably connected to the gear assembly **86**. The gear assembly **86** may be a speed-changing gear assembly configured to drive rotation of the second electric motor-generator **90** (e.g., the drive shaft **98**) at a different rotational speed than the rotational assembly **78**, and vice versa. The second electric motor-generator **90** is electrically connected to the electrical distribution system **92**. The second electric motor-generator **90** is configured to generate an electrical power output, in response to driving rotation from the rotational assembly **78** (e.g., the shaft **80**), and direct the electrical power output to the electrical distribution system **92** (e.g., a generating operational mode). Alternatively, the second electric motor-generator **90** is configured to receive an electrical power input from the electrical distribution system **92** (e.g., from the second electric motor-generator **90**, the energy storage device **94**, or a power source external to the propulsion system **20** or the aircraft **1000** (e.g., ground-based power system or an on-board power system of the aircraft **1000**) and drive or assist rotation of the rotational assembly **78** (e.g., the shaft **80**) through the gear assembly **86** (e.g., a motoring operational mode). The electrical power input and/or output of the second electric motor-generator **90** may be in the form of an alternating current (AC) or a direct current (DC), and the present disclosure is not limited to any particular configuration of the second electric motor-generator **90**.

[0048] The electrical distribution system **92** electrically connects components of the electrical assembly **26**. The electrical distribution system **92** includes switchgear, cables, wires, breakers, switches, electrical power conditional and/or conversion (e.g., AC to DC or DC to AC conversion) components, and/or other electrical components to effect the transfer of electrical power between components of the electrical assembly **26**. The electrical distribution system **92** may additionally include one or more electrical power controllers, for example, to control a magnitude and/or direction of electrical current flow to components of the electrical assembly **26**. The electrical distribution system **92** is configured to supply electrical power to one or more electrical aircraft loads **1002** of the aircraft **1000**. Examples of the electrical aircraft loads **1002** include, but are not limited to, electronic control systems, communication systems, and navigation systems, optical equipment, electrical actuators, and lasers.

[0049] The energy storage device **94** may be configured to store electrical energy (e.g., generated by the first electric motor-generator **88** and/or the second electric motor-generator **90**) for use by components of the electrical assembly **26** and/or the electrical aircraft loads **1002**. The energy storage device **94** may be configured as a battery, a capacitor (e.g., a supercapacitor), and/or another electrical energy storage device. For example, the energy storage device **94** may be configured as a battery including a plurality of battery modules (e.g., battery packs), battery cells, and/or the like electrically connected together in series and/or parallel as necessary to configure the battery with the desired electrical characteristics (e.g., voltage output, current output, storage capacity, etc.). The present disclosure is not limited to any particular battery configuration. For example, the battery (e.g., and its battery cells) may be configured as a rechargeable battery having

a battery chemistry such as, but not limited to, lead acid, nickel cadmium (NiCd), nickel-metal hydride (Ni-MH), lithium-ion (Li-ion), lithium-polymer (Li-poly), lithium metal, and the like.

[0050] The air intake **28** is configured to receive and direct ambient air to the cooling fan **30**. The air intake **28** may form a duct for the cooling fan **30** (e.g., within which duct the cooling fan **30** may be disposed). Alternatively, the air intake **28** may form a scoop or other opening at an exterior of the aircraft **1000** or its propulsion system **20**.

[0051] The cooling fan **30** includes a fan rotor **100**, an input shaft **102**, an air inlet **104**, and one or more air outlets **106**. The fan rotor **100** is connected to the input shaft **102**. The input shaft **102** is operably connected (e.g., coupled) to the gearbox **64** (e.g., the gear assembly **86**). The fan rotor **100** (e.g., the input shaft **102**) is configured to be rotationally driven by the rotational assembly **78** and/or the second electric motor generator **90** through the gear assembly **86**. The air inlet **104** is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the air intake **28**. The one or more air outlets **106** may be connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the intercooler **40** (e.g., the second inlet **58**) and the compressor **66** (e.g., the inlet **70**). The cooling fan **30** may additionally be configured to direct at least a portion of air flow from the air inlet **104** out of the aircraft **1000** and its propulsion system **20** from the one or more air outlets **106** (e.g., in a vertically downward direction) to generate lift for the aircraft **1000** (illustrated schematically in FIG. 2 as lifting airflow **108**).

[0052] Each of the propulsors **32** of FIG. 2 includes an electric motor **110** and a propulsor rotor **112**. The electric motor **110** is electrically connected to the electrical distribution system **92**. The electric motor **110** is operably connected (e.g., coupled) to the propulsor rotor **112** to drive rotation of the propulsor rotor **112**, for example, to provide lift, directional propulsion in three dimensions, and/or to control aircraft **1000** orientation (e.g., relative to a yaw axis, a pitch axis, and a roll axis). One or more of the propulsors **32** may be configured as a vertical take-off and landing (VTOL) propulsor. For example, one or more of the propulsors **32** may be pivotable to further control the aircraft **1000** lift, directional propulsion, and orientation. As shown in FIG. 1, for a quadcopter configuration of the aircraft **1000**, the propulsion system **20** may include four propulsors **32**. The present disclosure, however, is not limited to any particular number of the propulsors **32**. The propulsors **32** may alternatively be configured for short take-off and vertical landing (STOVL) or conventional take-off and landing operations. For example, the propulsors **32** may be configured as propeller (e.g., a tractor propeller configuration or a pusher propeller configuration).

[0053] In operation of the propulsion system **20**, ambient air is directed from the air intake **28** to the intercooler **40** (e.g., the second inlet **58**) and the compressor **66** (e.g., the inlet **70**) by the cooling fan **30**. As previously discussed, the cooling fan **30** may additionally direct air from the air intake **28** to generate lift for the aircraft **1000**. The ambient air is compressed by the compressor **66**, directed through the primary heat exchanger side of the intercooler **40** where the compressed air is cooled, and then supplied to the engine **34** (e.g., the air inlet **44**). The compressed air is mixed and burned with fuel in the engine **34** in an intermittent combustion cycle to drive rotation of the engine output shaft **42**. Combustion gas from the engine **34** is directed from the exhaust outlet **46** to the interburner **38** (e.g., the inlet **50**). This combustion gas from the engine **34** is mixed and burned with fuel in the interburner **38**, and the resultant combustion gas is directed from the outlet **52** to the inlet **74** and through the turbine **68**. The bladed turbine rotor **84** rotationally drives the rotational assembly **78** in response to the combustion gas flow through the turbine **68**. In particular, the bladed turbine rotor **84** rotationally drives the bladed compressor rotor **82**, to compress the ambient air for the engine **34**. The bladed turbine rotor **84** further rotationally drives the cooling fan **30** (e.g., the fan rotor **100** and the input shaft **102**) through the gearbox **64** (e.g., the gear assembly **86**). The propulsors **32** are powered by the electrical assembly **26** to provide lift, directional propulsion, and/or to control aircraft **1000** orientation.

[0054] During operation of the propulsion system **20**, the engine **34** (e.g., the engine output shaft **42**) may rotationally drive the first electric motor-generator **88** through the gearbox **36** (e.g., the

gear assembly **48**) to generate an electrical power output for use and/or storage by components of the electrical assembly **26** and/or the electrical aircraft loads **1002**. Similarly, the rotational assembly **78** (e.g., the shaft **80**) may rotationally drive the second electric motor-generator **90** through the gearbox **64** (e.g., the gear assembly **86**) to generate an electrical power output for use and/or storage by components of the electrical assembly **26** and/or the electrical aircraft loads **1002**. During some operating conditions, such as a propulsion system **20** startup sequence, the first electric motor-generator **88** may drive or assist rotation of the engine output shaft **42** using electrical power from the second electric motor-generator **90**, the energy storage device **94**, an onboard power supply (e.g., a generator and/or an energy storage device) of the aircraft **1000**, and/or a ground-based power supply **1008** electrically connected to the electrical distribution system **92**. Similarly, the second electric motor-generator **90** may drive or assist rotation of the rotational assembly **78** (e.g., the shaft) using electrical power from the first electric motor-generator **88**, the energy storage device **94**, an onboard power supply (e.g., a generator and/or an energy storage device) of the aircraft **1000**, and/or a ground-based power supply electrically connected to the electrical distribution system **92**.

[0055] The present disclosure configuration of the propulsion system **20** facilitates mechanical independence between the engine **34** (e.g., the engine output shaft **38**) and the turbocompressor **62** (e.g., the rotational assembly **78**). In other words, the engine output shaft **42** and the rotational assembly **78** are not operably connected by any mechanical coupling component or assembly such as, but not limited to, a shaft, a gearbox, a belt, a clutch, a transmission (e.g., a hydrostatic transmission), or the like. This mechanical independence between the engine **34** and the turbocompressor **62** facilitates greater flexibility in propulsion system **20** packaging (e.g., relative orientations and proximity between the engine **34** and the turbocompressor **62**) within the aircraft **1000**. This mechanical independence between the engine **34** and the turbocompressor **62** additionally facilitates independent speed control of the engine **34** and the turbocompressor **62**, thereby facilitating optimal operation across a range of aircraft **1000** altitudes and operating conditions. For the propulsion system **20**, the engine **34** and the turbocompressor **62** independently facilitate generation of electrical power for the components of the electrical assembly **26** and/or the electrical aircraft loads **1002**. For example, independent fuel flows to the engine **34** and the interburner **38** facilitate optimized turbocompressor **62** speed and engine **34** power for aircraft **1000** propulsion while efficiently meeting electrical loading (e.g., electrical aircraft loads **1002**) requirements, thereby providing significant fuel savings in comparison to conventional intermittent combustion engine or gas turbine engine configurations.

[0056] Referring to FIG. 3, in some embodiments of the propulsion system **20**, the exhaust outlet **46** may be directly connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the inlet **74**. In other words, the propulsion system **20** may not include the interburner **38** (see FIG. 2). In operation of the propulsion system **20** of FIG. 3, the combustion gas from the engine **34** is directed from the exhaust outlet **46** to the inlet **74** and through the turbine **68**. The bladed turbine rotor **84** rotationally drives the rotational assembly **78** in response to the combustion gas flow through the turbine **68**. In particular, the bladed turbine rotor **84** rotationally drives the bladed compressor rotor **82**, to compress the ambient air for the engine **34**. The bladed turbine rotor **84** further rotationally drives the cooling fan **30** (e.g., the fan rotor **100** and the input shaft **102**) through the gearbox **64** (e.g., the gear assembly **86**). The direct fluid connection between the exhaust outlet **46** and the inlet **74** may facilitate a simplified propulsion system **20** configuration and/or alternative packaging configurations of the propulsion system **20** within the aircraft **1000** (see FIG. 1).

[0057] Referring to FIG. 4, in some embodiments of the propulsion system **20**, the electrical assembly **26** may not include an electric motor and/or an electric generator (e.g., the second electric motor-generator **90**; see FIG. 2) mechanically coupled with the rotational assembly **78** or the cooling fan **30** (e.g., through the gearbox **64** and its gear assembly **86**). For example, in operation

of the propulsion system **20** of FIG. **4**, the fan rotor **100** may be driven solely by the rotational assembly **78** (e.g., the shaft **80**) through the gearbox **64** (e.g., the gear assembly **86**). The fan rotor **100** driven only by the rotational assembly **78** may facilitate a simplified propulsion system **20** configuration and/or alternative packaging configurations of the propulsion system **20** within the aircraft **1000**.

[0058] Referring to FIG. **5**, in some embodiments of the propulsion system **20**, the compressor **66** (e.g., the inlet **70**) may be directly connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the air intake **28**. For example, in operation of the propulsion system **20** of FIG. **5**, ambient air may be directed from the air intake **28** to the compressor **66** (e.g., the inlet **70**). The ambient air from the air intake **28** may be compressed by the compressor **66**, directed through the primary heat exchanger side of the intercooler **40** where the compressed air is cooled, and then supplied to the engine **34** (e.g., the air inlet **44**). The direct fluid connection between the air intake **28** and the compressor **66** may facilitate alternative packaging configurations of the propulsion system **20** within the aircraft **1000** (see FIG. **2**) and/or facilitate improved generation of the lift by the cooling fan **30**.

[0059] Referring to FIG. **6**, in some embodiments of the propulsion system **20**, the cooling fan **30** (e.g., the air inlet **104**) may be connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with and downstream of the intercooler **40** (e.g., the second outlet **60**). The cooling air from the intercooler **40** is exhausted from the cooling fan **30** through the air outlet **106** (e.g., as the lifting airflow **108**). The electrical assembly **26** of FIG. **6** includes an electric motor **114**. The electric motor **114** is operably connected (e.g., coupled) to the fan rotor **100** (e.g., the input shaft **102**) to drive rotation of the fan rotor **100**. The electric motor **114** is electrically connected to the electrical distribution system **92**. In operation of the propulsion system **20** of FIG. **6**, ambient air from the air intake **28** may be drawn through the secondary heat exchanger side of the intercooler **40** (e.g., from the second inlet **58** to the second outlet **60**) and exhausted from the aircraft **1000** (see FIG. **1**) and its propulsion system **20** through the air outlet **106**. The exhaust of the air from the air outlet **106** may generate additional lift for the aircraft **1000**. The configuration of the cooling fan **30** at the second outlet **60** may facilitate improved cooling air flow through the second heat exchanger side of the intercooler **40**, improved generation of the lift by the cooling fan **30**, and alternative packaging configurations of the propulsion system **20** within the aircraft **1000**.

[0060] Referring to FIG. **7**, in some embodiments of the propulsion system **20**, the air intake **28** may be directly connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the intercooler **40** (e.g., the second inlet **58**) and the compressor **66** (e.g., the inlet **70**). In other words, the propulsion system **20** may not include the cooling fan **30** (see FIG. **2**). The intercooler **40** (e.g., the second outlet **60**) is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with one or more of the propulsors **32**. In operation of the propulsion system **20** of FIG. **7**, ambient air is directed from the air intake **28** to the intercooler **40** (e.g., the second inlet **58**) and the compressor **66** (e.g., the inlet **70**). The ambient air is drawn through the second heat exchanger side of the intercooler **40** by the propulsors **32**. The propulsors **32** direct the ambient air from the intercooler **40** to provide lift, directional propulsion, and/or to control aircraft **1000** orientation (see FIG. **1**). As shown in FIG. **8**, one or more of the propulsors **32** may have a ducted configuration in which the propulsor rotor **112** is circumscribed by a propulsor duct **116**. The propulsor duct **116** may form a propulsor inlet **118** and a propulsor outlet **120**. The propulsor inlet **118** may be connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with the intercooler **40** (e.g., the second outlet **60**). Rotation of the propulsor rotor **112** (e.g., by the electric motor **114**) within the propulsor duct **116** draws air through the intercooler **40** and directs the air from the propulsor outlet **120** to provide lift, directional propulsion, and/or to control aircraft **1000** orientation. The ambient air is compressed by the compressor **66**, directed through the primary heat exchanger side of the intercooler **40** where the compressed air is cooled, and then supplied to the engine **34** (e.g., the air inlet **44**).

[0061] Referring to FIG. 9, in some embodiments of the propulsion system 20, the electrical assembly 26 may include an electric motor 122. The electric motor 122 is operably connected (e.g., coupled) to the fan rotor 100 (e.g., the input shaft 102) to drive rotation of the fan rotor 100. The electric motor 122 is electrically connected to the electrical distribution system 92. In operation of the propulsion system 20 of FIG. 9, ambient air from the air intake 28 is directed by the cooling fan 30 to the intercooler 40 (e.g., the second inlet 58) and the compressor 66 (e.g., the inlet 70). The cooling fan 30 may additionally exhaust air from the air outlet 106 to generate additional lift for the aircraft 1000. Additionally or alternatively, in some embodiments of the propulsion system 20, the turbocompressor 62 may be configured as a single module with the second electric motor-generator 90. The rotational assembly 78 may be directly coupled with the second electric motor-generator 90. For example, the shaft 80 may drive rotation of the second electric motor-generator 90 at a same rotational speed as the other rotational components of the rotational assembly 78. As a result, the turbocompressor assembly 24 of the propulsion system 20 of FIG. 9 may not include the gearbox 64 (see, e.g., FIG. 2). The configuration of the turbocompressor 62, the second electric motor-generator 90, and the cooling fan 30 of the propulsion system 20 of FIG. 9 facilitates reduced mechanical complexity of the propulsion system 20 as well as reduced weight and manufacturing cost.

[0062] Referring to FIG. 10, in some embodiments of the propulsion system 20, the propulsion system 20 may include an engine exhaust diverter assembly 124. The engine exhaust diverter assembly 124 is connected in fluid communication (e.g., using any suitable conduit, pipe, tube, hose, etc.) with and between the engine 34 (e.g., the exhaust outlet 46), the interburner 26 (e.g., the inlet 50), and the turbine 68 (e.g., the inlet 74). The engine exhaust diverter subassembly 124 is a multi-directional flow diverter. In a first condition, the engine exhaust diverter assembly 124 may direct combustion gas from the exhaust outlet 46 to the inlet 50. In this first condition, the engine exhaust diverter assembly 124 may prevent or substantially prevent the flow of the combustion gas from (e.g., directly from) the exhaust outlet 46 to the turbine 68 (e.g., the inlet 74). In a second condition, the engine exhaust diverter assembly 124 may direct combustion gas from the exhaust outlet 46 to the turbine 68 (e.g., the inlet 74). In this second condition, the engine exhaust diverter assembly 124 may prevent or substantially prevent the flow of the combustion gas from (e.g., directly from) the exhaust outlet 46 to the inlet 50. For example, during higher-power conditions of the propulsion system 20, the engine exhaust diverter assembly 124 may be configured in the first condition to direct the combustion gas from the exhaust outlet 46 to the inlet 50 to be mixed and burned with fuel in the interburner 38, and then subsequently directed from the interburner 38 (e.g., the outlet 52) to the turbine 68 (e.g., the inlet 74). During lower-power conditions of the propulsion system 20, the engine exhaust diverter assembly 124 may be configured in the second condition to direct the combustion gas from the exhaust outlet 46 to the inlet 74, thereby bypassing the interburner 38. The engine exhaust diverter assembly 124 may also be configurable in a closed condition in which the combustion gas from the exhaust outlet 46 does not flow through the engine exhaust diverter assembly 124 (e.g., to the interburner 38 or the turbine 68). In other words, in the closed condition, the engine exhaust diverter assembly 124 isolates the exhaust outlet 46 from the inlet 50 and the inlet 74. The engine exhaust diverter assembly 124 may be formed by a remotely-actuated multi-directional valve or an assembly of remotely-actuated valves configured to selectively direct the combustion gas from the exhaust outlet 46 as described above. The present disclosure, however, is not limited to the foregoing exemplary valve configuration of the engine exhaust diverter assembly 124.

[0063] While the principles of the disclosure have been described above in connection with specific apparatuses and methods, it is to be clearly understood that this description is made only by way of example and not as limitation on the scope of the disclosure. Specific details are given in the above description to provide a thorough understanding of the embodiments. However, it is understood that the embodiments may be practiced without these specific details.

[0064] It is noted that the embodiments may be described as a process which is depicted as a flowchart, a flow diagram, a block diagram, etc. Although any one of these structures may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc.

[0065] The singular forms “a,” “an,” and “the” refer to one or more than one, unless the context clearly dictates otherwise. For example, the term “comprising a specimen” includes single or plural specimens and is considered equivalent to the phrase “comprising at least one specimen.” The term “or” refers to a single element of stated alternative elements or a combination of two or more elements unless the context clearly indicates otherwise. As used herein, “comprises” means “includes.” Thus, “comprising A or B,” means “including A or B, or A and B,” without excluding additional elements.

[0066] It is noted that various connections are set forth between elements in the present description and drawings (the contents of which are included in this disclosure by way of reference). It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. Any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option.

[0067] No element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprise”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

[0068] While various inventive aspects, concepts and features of the disclosures may be described and illustrated herein as embodied in combination in the exemplary embodiments, these various aspects, concepts, and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the present application. Still further, while various alternative embodiments as to the various aspects, concepts, and features of the disclosures—such as alternative materials, structures, configurations, methods, devices, and components, and so on—may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed. Those skilled in the art may readily adopt one or more of the inventive aspects, concepts, or features into additional embodiments and uses within the scope of the present application even if such embodiments are not expressly disclosed herein. For example, in the exemplary embodiments described above within the Detailed Description portion of the present specification, elements may be described as individual units and shown as independent of one another to facilitate the description. In alternative embodiments, such elements may be configured as combined elements.

Claims

1. A propulsion system for an aircraft, the propulsion system comprising: an air intake; an engine assembly including an engine and an intercooler, the engine including an air inlet, an exhaust outlet, and an engine output shaft, and the intercooler including a first heat exchanger side having a first inlet and a first outlet and a second heat exchanger side having a second inlet and a second outlet, the first outlet is connected in fluid communication with the air inlet, and the second inlet is

connected in fluid communication with the air intake; a turbocompressor assembly including a turbocompressor, the turbocompressor including a turbine and a compressor, the turbine and the compressor forming a rotational assembly, the rotational assembly including a shaft, a bladed turbine rotor of the turbine, and a bladed compressor rotor of the compressor, the shaft interconnecting the bladed turbine rotor and the bladed compressor rotor, the turbine including a turbine inlet and a turbine outlet, the turbine inlet is connected in fluid communication with the exhaust outlet, the compressor including a compressor inlet and a compressor outlet, the compressor inlet is connected in fluid communication with the air intake, the compressor outlet is connected in fluid communication with the first inlet, and the rotational assembly is mechanically independent of the engine output shaft; an electrical assembly including a first electric motor-generator and an electrical distribution system, the first electric motor-generator operably connected to the engine output shaft, the first electric motor-generator electrically connected to the electrical distribution system; and at least one propulsor including an electric motor and a propulsor rotor, the electric motor is electrically connected to the electrical distribution system, and the propulsor rotor is operably connected to the electric motor.

2. The propulsion system of claim 1, further comprising a cooling fan including a fan rotor and an input shaft, the input shaft is connected to the fan rotor, the input shaft is operably connected to the rotational assembly, the cooling fan including a cooling fan inlet and at least one cooling fan outlet, and the cooling fan is connected in fluid communication with the second heat exchanger side to direct ambient air from the air intake through the second heat exchanger side.

3. The propulsion system of claim 2, wherein the cooling fan is connected in fluid communication with and between the air intake and the second heat exchanger side with the cooling fan inlet connected in fluid communication with the air intake and the at least one cooling fan outlet connected in fluid communication with the second inlet.

4. The propulsion system of claim 2, wherein the cooling fan is configured to exhaust ambient air from the air intake through the at least one cooling fan outlet to generate lift for the aircraft.

5. The propulsion system of claim 2, wherein the electrical assembly includes a second electric motor-generator operably connected to the shaft, and the second electric motor-generator is electrically connected to the electrical distribution system.

6. The propulsion system of claim 5, wherein the turbocompressor assembly further includes a gearbox, and the gearbox operably connects the shaft, the input shaft, and the second electric motor-generator.

7. The propulsion system of claim 1, further comprising a cooling fan including an electric motor, a fan rotor, a cooling fan inlet, and at least one cooling fan outlet, the electric motor is electrically connected to the electrical distribution system, the electric motor is operably connected to the fan rotor, and the cooling fan is connected in fluid communication with the second heat exchanger side to direct ambient air from the air intake through the second heat exchanger side.

8. The propulsion system of claim 7, wherein the cooling fan is connected in fluid communication with and downstream of the second heat exchanger side, and the cooling fan inlet is connected in fluid communication with the second outlet.

9. The propulsion system of claim 1, wherein the engine assembly includes an interburner, the interburner includes an interburner inlet and an interburner outlet, the interburner is connected in fluid communication with and between the engine and the turbine with the interburner inlet connected in fluid communication with the exhaust outlet and the interburner outlet connected in fluid communication with the turbine inlet.

10. The propulsion system of claim 1, wherein the electrical assembly further includes an energy storage device electrically connected to the electrical distribution system.

11. The propulsion system of claim 1, wherein each of the at least one propulsor is connected in fluid communication with the intercooler outlet.

12. The propulsion system of claim 1, wherein the engine is an intermittent combustion engine.

13. An unmanned aerial vehicle (UAV) comprising: an air intake; a turbocompressor assembly including a turbocompressor, the turbocompressor including a turbine and a compressor, the turbine and the compressor forming a rotational assembly, the rotational assembly including a shaft, a bladed turbine rotor of the turbine, and a bladed compressor rotor of the compressor, the shaft interconnecting the bladed turbine rotor and the bladed compressor rotor, the turbine including a turbine inlet and a turbine outlet, the compressor including a compressor inlet and a compressor outlet; a cooling fan including a fan rotor and an input shaft, the input shaft is connected to the fan rotor, the input shaft is operably connected to the rotational assembly, the cooling fan including a cooling fan inlet and at least one cooling fan outlet, the cooling fan inlet is connected in fluid communication with the air intake, and the cooling fan outlet is connected in fluid communication with the compressor inlet; and an engine assembly including an engine and an intercooler, the engine including an air inlet, an exhaust outlet, and an engine output shaft, and the exhaust outlet is connected in fluid communication with the turbine inlet, and the intercooler including a first heat exchanger side having a first inlet and a first outlet and a second heat exchanger side having a second inlet and a second outlet, the first inlet is connected in fluid communication with the compressor outlet, the first outlet is connected in fluid communication with the air inlet, and the second inlet is connected in fluid communication with the cooling fan outlet.

14. The UAV of claim 13, further comprising an electrical assembly including a first electric motor-generator, a second electric motor-generator, and an electrical distribution system, the first electric motor-generator is operably connected to the engine output shaft, the first electric motor-generator is electrically connected to the electrical distribution system, the second electric motor-generator is operably connected to the rotational assembly, and the second electric motor-generator is electrically connected to the electrical distribution system.

15. The UAV of claim 14, wherein the input shaft is operably connected to the rotational assembly and the second electric motor-generator.

16. The UAV of claim 14, further comprising at least one propulsor including an electric motor and a propulsor rotor, the electric motor is electrically connected to the electrical distribution system, and the propulsor rotor is operably connected to the electric motor.

17. A propulsion system for an aircraft, the propulsion system comprising: an air intake; an engine assembly including an engine, an interburner, and an intercooler, the engine including an air inlet, an exhaust outlet, and an engine output shaft, the interburner including an interburner inlet and an interburner outlet, the interburner inlet is connected in fluid communication with the exhaust outlet, and the intercooler including a first heat exchanger side having a first inlet and a first outlet and a second heat exchanger side having a second inlet and a second outlet, the first outlet is connected in fluid communication with the air inlet, and the second inlet is connected in fluid communication with the air intake; a turbocompressor assembly including a turbocompressor, the turbocompressor including a turbine and a compressor, the turbine and the compressor forming a rotational assembly, the rotational assembly including a shaft, a bladed turbine rotor of the turbine, and a bladed compressor rotor of the compressor, the shaft interconnecting the bladed turbine rotor and the bladed compressor rotor, the turbine including a turbine inlet and a turbine outlet, the turbine inlet is connected in fluid communication with the interburner outlet, the compressor including a compressor inlet and a compressor outlet, the compressor inlet is connected in fluid communication with the air intake, the compressor outlet is connected in fluid communication with the first inlet, and the rotational assembly is mechanically independent of the engine output shaft; and an electrical assembly including a first electric motor-generator, a second electric motor-generator, and an electrical distribution system, the first electric motor-generator is operably connected to the engine output shaft, the first electric motor-generator is electrically connected to the electrical distribution system, the second electric motor-generator is operably connected to the rotational assembly, and the second electric motor-generator is electrically connected to the electrical distribution system.

18. The propulsion system of claim 17, further comprising a cooling fan including a fan rotor and an input shaft, the input shaft is connected to the fan rotor, the input shaft is operably connected to the rotational assembly, the cooling fan including a cooling fan inlet and at least one cooling fan outlet, the cooling fan inlet is connected in fluid communication with the air intake, and the at least one cooling fan outlet is connected in fluid communication with the compressor inlet and the second inlet.

19. The propulsion system of claim 18, wherein the cooling fan is configured to exhaust ambient air from the air intake through the at least one cooling fan outlet to generate lift for the aircraft.

20. The propulsion system of claim 17, further comprising an engine exhaust diverter assembly connected in fluid communication with and between the exhaust outlet, the interburner inlet, and the turbine inlet, and the engine exhaust diverter assembly is selectively configurable in a first condition or a second condition: in the first condition, the engine exhaust diverter subassembly connects the exhaust outlet in fluid communication with the interburner inlet, and in the second condition, the engine exhaust diverter subassembly connects the exhaust outlet in fluid communication with the turbine inlet.
